

Notes: Zingales (JF, 1998)

Survival of the Fittest or the Fattest? Exit and Financing in the Trucking Industry

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1 Big picture

- **Question.** Does financing affect which firms survive a major product-market shock? The paper asks whether financially weak but potentially efficient trucking firms were more likely to exit after trucking deregulation.
- **Core idea.** Standard evolutionary reasoning says competition selects the most efficient firms. Zingales asks whether capital-market imperfections distort that selection process so that the “fattest” firms, meaning firms with stronger balance sheets, survive alongside or instead of the “fittest” firms.
- **Setting.** The empirical setting is the U.S. interstate trucking industry around the Carter-era deregulation and the Motor Carrier Act of 1980. Deregulation sharply increased price competition and entry, and it destroyed much of the value of operating certificates that had functioned like monopoly licenses.
- **Identification intuition.** Prederegulation leverage was chosen under a regulated regime. Deregulation was largely unexpected and changed both competition and the effective leverage of carriers by reducing the value of operating rights. This makes 1976–1977 leverage a useful source of variation for studying how financing shaped survival during the industry shock.
- **Main result.** Highly leveraged trucking firms were significantly less likely to survive until 1985, even after controlling for profitability, size, intangibles, labor cost, region, alternative efficiency measures, and Altman’s Z-score default-risk measure.
- **Mechanism result.** Debt appears to matter partly through underinvestment and pricing. More highly leveraged carriers invested less early in deregulation, especially among eventual exiters, and charged lower prices during the postderegulation price war.
- **Heterogeneity result.** The negative survival effect of leverage is concentrated in the less-than-truckload (LTL) segment, where networks, service quality, and imperfect competition are more important. It is absent in the more competitive truckload (TL) segment.
- **Economic takeaway.** Product-market competition alone may not select only the most efficient firms. Financial structure can affect survival, investment, and pricing, so industry shakeouts can select firms partly by financial robustness.

2 Incremental contribution of the paper

- **From theory to empirics.** Theoretical corporate-finance models had already argued that debt overhang, predation, and product-market interactions can make financing matter for real decisions. This paper provides a direct empirical test in a real industry shock.
- **Natural experiment contribution.** Instead of studying leverage choices that may be endogenously chosen for strategic reasons, the paper studies leverage before an unexpected deregulation shock. This helps separate the effects of financing from normal capital-structure choice.
- **Selection contribution.** The paper reframes survival analysis. It asks whether exit reflects only productive efficiency or also balance-sheet strength.

- **Industry contribution.** The trucking industry is ideal because regulation created valuable operating certificates and muted price competition; deregulation suddenly destroyed these rents and intensified competition.
- **Mechanism contribution.** The paper does not only regress exit on leverage. It also studies investment and prices to understand how debt may reduce survival.
- **Market-structure contribution.** By comparing TL and LTL segments, the paper links the financing-survival channel to the degree of product-market imperfection and asset tangibility.
- **Broader corporate-finance contribution.** The findings help connect capital structure to product-market outcomes, complementing evidence from leveraged buyouts but using a shock that is less likely to be chosen by firms themselves.

3 Institutional background: trucking regulation and deregulation

3.1 Before deregulation

- Interstate motor carriers were regulated under the Motor Carrier Act of 1935.
- The Interstate Commerce Commission (ICC) controlled entry and rates. Carriers needed operating certificates to serve routes.
- Operating certificates were valuable because they represented protected route rights. These certificates were economically similar to monopoly licenses.
- Regulation limited entry and price competition, generating rents for certificate holders and for unionized labor.
- Because operating rights were valuable, carriers' book leverage understated their true economic equity value before deregulation. Once operating rights lost value, effective leverage rose sharply.

3.2 Deregulation shock

- The deregulation process began with policy changes at the ICC in the late 1970s and was codified by the Motor Carrier Act of 1980.
- New entry became much easier and price competition intensified.
- The stock-market value of publicly traded trucking companies fell sharply around the deregulation period, consistent with the view that the shock destroyed regulated rents.
- Between 1980 and 1985, the industry experienced a large shakeout with many exits and many new entrants.

Interpretation: The key empirical benefit is that deregulation simultaneously changed the product market and shocked firms' balance sheets. Firms entered the shock with capital structures chosen for the old regime, not for the new competitive environment.

3.3 Truckload versus less-than-truckload carriers

- **Truckload (TL).** TL carriers haul larger shipments. The segment is closer to competitive: assets are mostly trucks, capital is easier to collateralize, entry is easier, and service is more standardized.
- **Less-than-truckload (LTL).** LTL carriers haul smaller shipments and rely on networks of terminals, hubs, and coordination. The segment has scale economies, more intangible network value, stronger customer concerns about service reliability, and less perfect competition.
- The paper predicts that debt should matter more in LTL if debt affects investment, service reputation, customer confidence, or exposure to predation.

4 Theoretical hypotheses

4.1 Why leverage may reduce survival

- **Debt overhang and underinvestment.** Highly levered firms may pass up positive-NPV investments because the gains accrue partly to creditors. This is especially plausible in LTL, where investment includes network and terminal assets that are less easily collateralized.
- **Predation and deep pockets.** In models such as Bolton and Scharfstein, competitors can prey on financially weak firms. This channel should be most relevant in markets with imperfect competition and some barriers to entry.
- **Customer and reputation effects.** Customers may avoid financially weak carriers if bankruptcy would disrupt shipments. Reputation and reliability are more important in LTL than in standardized TL hauling.
- **Financial fragility.** High leverage may increase bankruptcy risk mechanically, but the paper tries to separate this channel from a product-market channel by controlling for Altman's Z-score.

4.2 Why leverage may increase survival

- **Efficient liquidation.** Debt can force inefficient firms to exit. If this is the mechanism, the negative association between leverage and survival is not a cost of debt but a benefit.
- **Hard restructuring.** Debt can force firms to renegotiate with workers, cut costs, and restructure earlier.
- **Strategic aggression.** Leveraged equity resembles a call option; highly levered firms might compete more aggressively and survive if aggressive expansion is the winning strategy.

Interpretation: The sign of the leverage-survival relationship is therefore theoretically ambiguous. The paper's contribution is to test which forces dominate in a setting where leverage was shocked by deregulation.

5 Data and measurement

5.1 Data source and sample

- The data come from ICC filings collected and reclassified by the American Trucking Association (ATA).
- The period is 1976 to 1985.
- Carriers disclose financial variables and operating variables such as shipments, ton-miles, average haul, average load, and revenue per ton-mile.
- The analysis focuses on **general freight carriers**, the largest market segment and the group with richer information.
- The core sample consists of Class I and II general freight carriers with more than \$1 million in operating revenues in 1977.
- Prederegulation variables are usually measured as the average of 1976 and 1977 values.

5.2 Exit definition

- A carrier is treated as exiting when it disappears from the ATA data set and does not reappear in any later year through 1985.
- Exit includes liquidation, bankruptcy, and acquisition followed by merger into another carrier.
- The measure captures survival as a separate trucking organization rather than survival of physical trucks or route assets.
- The revenue cutoff is designed to avoid classifying temporary drops below reporting thresholds as exits.

5.3 Key variables

- **Leverage.** The main leverage variable is net debt-to-capital: total debt minus cash reserves, divided by debt plus book equity.
- **Efficiency.** The main operating efficiency/profitability proxy is return on sales (EBITDA divided by operating revenues). The paper also uses return on assets, estimated technical inefficiency, cost per ton-mile, and 1980 return on sales.
- **Prederegulation monopoly rents.** Intangibles over total assets proxies for operating-certificate value and monopoly rents.
- **Labor cost and unionization.** Wages and benefits divided by operating costs proxies for labor intensity, purchased transportation, and union exposure.
- **Market segment.** The fraction of revenues from less-than-truckload shipments divides carriers into low, medium, and high LTL groups.
- **Default risk.** Altman's Z-score is used as a proxy for the ex ante probability of default.

6 Empirical specifications

6.1 Survival regressions

The baseline survival regression asks whether leverage predicts survival after controlling for operating characteristics:

$$\Pr(\text{Survival}_{i,1985} = 1) = F(X_{i,1977}, \text{Lev}_{i,1977}) + \varepsilon_i.$$

The vector $X_{i,1977}$ includes profitability, size, intangibles, labor cost, and regional dummies. The leverage variable is measured before the deregulation shock and before most of the exit wave.

6.2 Interpreting the leverage coefficient

- A negative coefficient on leverage means more debt reduces the probability of survival, conditional on observed efficiency and other controls.
- The key concern is that leverage may proxy for unobserved poor quality. The paper responds by adding multiple efficiency measures, using 1980 profitability, controlling for default risk, and comparing TL/LTL segments.
- The segment comparison is especially important. If omitted quality drove the result in the same way across all carriers, leverage should matter similarly across TL and LTL. Instead, the effect is concentrated in the LTL segment.

6.3 Investment and pricing regressions

To study mechanisms, the paper estimates investment and pricing equations. Investment is measured as the change in gross operating property over a period divided by 1977 net operating property. Prices are proxied by revenue per ton-mile and are studied as changes relative to 1977.

Interpretation: These mechanism regressions ask whether leverage constrains real adjustment after deregulation: Does high debt reduce investment? Does it force firms to cut prices during the price war?

7 Tables and discussion

7.1 Table I: Industry Segments in 1977 and Table II: Summary Statistics

Table I
Industry Segments in 1977

This table reports the distribution by main activity (types of goods hauled) of the 3,150 carriers reporting to the Interstate Commerce Commission (ICC) in 1977. These categories can be thought of as different market segments. The data used come from the American Trucking Association, which has been collecting and reclassifying ICC filings since 1976. In the rest of the paper, I focus only on general freight carriers.

Commodity	Frequency	Percent
General freight	1,300	41.27
Household goods	248	7.87
Heavy machinery	80	2.54
Petroleum products	162	5.14
Refrigerated liquids	12	0.38
Refrigerated solids	143	4.54
Dump trucking	84	2.67
Agricultural commodity	97	3.08
Motor vehicles	49	1.56
Armored truck service	2	0.06
Building materials	135	4.29
Films & Associated commodities	5	0.16
Forest products	22	0.70
Mine ores not including coal	5	0.16
Retail store delivery service	26	0.83
Dangerous and hazardous materials	7	0.22
Other commodities not elsewhere classified	773	24.54
Total	3,150	100.00

Table II
Summary Statistics

Data are presented for the population of 941 general freight carriers with more than \$1 million in operating revenues in 1977. Variables are defined in Panel A and are averages of the 1976 and 1977 values. Advance payables include any notes payable. Negative book values of equity are set equal to zero. Coverage is set equal to zero when earnings before interest, taxes, and depreciation (EBITDA) are less than or equal to zero and is set equal to 100 when coverage is greater than or equal to 100. Observations with a return on assets larger than 1 or smaller than -1 are set equal to missing. The difference column (Diff.) reports the difference in means between carriers that were still alive in 1985 and carriers that were not alive at that date. The p -values refer to the null hypothesis that the difference between the two means is equal to zero.

Panel A: Variable Definitions

Variable	Definition
Average load	Ton-miles/Total miles
Average haul	Ton-miles hauled/Tons of revenue freight
Cost per ton-mile	Operating expenses/Ton-miles hauled
Coverage	EBITDA/Interest expenses
Debt	Short term + Long term debt + Advances payables
Debt-to-capital	Debt/(Debt + Equity)
Intangibles	Intangibles/Total assets
Labor cost	(Wages + Benefits)/Operating costs
Net debt-to-capital	(Debt - Cash reserves)/(Debt + Equity)
Market share	Operating revenues/Total operating revenues of carriers in the same state
Proportion LTL revenues	Freight revenues up to 10,000 pounds/Freight revenues
Return on assets	EBITDA/Total assets
Return on sales	EBITDA/Operating revenues
Revenue per ton-mile	Operating revenues/Ton-miles hauled

Panel B: Summary Statistics

Variable	Mean	Median	Std. Dev.	Min.	Max.	Diff.	p -value	Obs.
Revenues (\$ million)	15.0	3.25	47.52	0.90	684	6.33	0.041	941
Assets (\$ million)	6.88	1.22	2.41	0.05	355	4.08	0.009	941
Debt-to-capital	0.40	0.35	0.30	0.00	1.00	-0.10	0.000	940
Net debt-to-capital	0.22	0.20	0.38	-0.97	1.00	-0.13	0.000	910
Coverage	27.33	9.78	33.42	0.00	100	10.96	0.000	923
Return on assets	0.18	0.18	0.13	-0.45	0.88	0.07	0.000	939
Return on sales	0.08	0.07	0.06	-0.43	0.52	0.04	0.000	941
Intangibles	0.06	0.03	0.08	0.00	0.64	-0.01	0.005	930
Labor cost	0.48	0.54	0.18	0.01	0.88	-0.03	0.005	939
Revenue per ton-mile	0.32	0.21	0.42	0.02	6.9	-0.03	0.405	773
Cost per ton-mile	0.41	0.21	0.70	0.00	6.87	-0.09	0.059	768
Proportion LTL revenues	0.58	0.66	0.27	0.00	1.00	0.00	0.949	649
Average load	10.32	10.04	5.33	0.47	36.54	0.26	0.502	772
Average haul	227.15	169.40	209.92	6.93	1713	-27.70	0.105	606
Market share	0.05	0.01	0.12	0.00	1.00	0.02	0.016	941
Number of employees	418.46	86.25	1405.37	2.00	26,035	184.36	0.048	906

(a) Table I: Industry Segments in 1977

(b) Table II: Summary Statistics

Read: Table I shows that general freight is the largest trucking segment, accounting for 1,300 of 3,150 carriers, or 41.27% of the sample. This motivates focusing on general freight carriers. Table II defines key variables and summarizes the main sample of 941 general freight carriers with more than \$1 million in 1977 revenues.

Economic message: Table II already previews the main empirical pattern: surviving firms are larger, more profitable, less levered, have higher coverage, and have lower labor-cost shares. The raw data therefore support both the “fittest” channel and the “fattest” channel.

7.2 Table III: Effect of Initial Leverage on the Probability of Survival and Table IV: Robustness to Different Measures of Efficiency

Table III

Effect of Initial Leverage on the Probability of Survival

The dependent variable is the probability that a general freight motor carrier with more than \$1 million in revenues in 1977 survives until 1985. The reported coefficients are probit estimates of the effect of a marginal change in the corresponding regressor on the probability of survival, computed at the average of the dependent variable. All the independent variables are measured as of 1977 (average of the 1976 and 1977 values). Net debt-to-capital is total debt minus cash reserves divided by total debt plus equity. Intangibles is the ratio of intangible assets to total assets. Labor cost is the ratio of the wages and benefits to operating costs. Among the independent variables there are also nine regional dummies (not reported). Standard errors are reported in parentheses.

Independent Variables	I	II	III
Net debt-to-capital	-0.180*** (0.049)	-0.180*** (0.048)	
log(1 + Coverage)			0.042*** (0.016)
Return on sales	2.878*** (0.347)		2.584*** (0.363)
Return on assets		0.925*** (0.134)	
log(Revenues)	0.048*** (0.014)	0.044*** (0.014)	0.043*** (0.014)
Intangibles	-0.299 (0.249)	-0.583** (0.245)	-0.355 (0.245)
Labor cost	-0.324*** (0.103)	-0.233** (0.103)	-0.319*** (0.102)
Pseudo- R^2	0.12	0.09	0.11
N	889	887	902

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

(a) Table III: Effect of Initial Leverage on the Probability of Survival

Table IV

Robustness to Different Measures of Efficiency

The dependent variable is the probability that a general freight motor carrier with more than \$1 million in revenues in 1977 survives until 1985. The reported coefficients are probit estimates of the effect of a marginal change in the corresponding regressor on the probability of survival, computed at the average of the dependent variable. Net debt-to-capital is total debt minus cash reserves divided by total debt plus equity. Intangibles is the ratio of intangible assets to total assets. Labor cost is the ratio of the wages and benefits to operating costs. All the independent variables are measured as of 1977 (average of the 1976 and 1977 values) except the 1980 return on sales in the last column. Column I uses the estimated inefficiency of a carrier as additional proxy for a carrier's efficiency. This is computed as the distance of its output from the production frontier, estimated using a translog function (see the Appendix). Column II uses as additional proxy for efficiency the cost per ton-mile. Cost per ton-mile is the ratio of operating expenses to ton-miles hauled. Column III uses the ex post value of efficiency as of 1980 as a proxy for efficiency. In such case the dependent variable is the probability of survival after 1980. All regressions contain nine regional dummies (not reported). Standard errors are reported in parentheses.

Independent Variables	I	II	III
Net debt-to-capital	-0.245*** (0.066)	-0.226*** (0.057)	-0.277*** (0.058)
Return on sales	3.142*** (0.481)	3.101*** (0.410)	1.131** (0.512)
log(Revenues)	0.030* (0.017)	0.029* (0.016)	0.022 (0.015)
Intangibles	-0.045 (0.303)	-0.234 (0.274)	-0.066 (0.280)
Labor cost	-0.332** (0.152)	-0.436*** (0.124)	-0.187 (0.119)
Estimated inefficiency	0.504* (0.266)		
Cost per ton-mile		-0.019 (0.032)	
Return on sales in 1980			2.450*** (0.452)
Pseudo- R^2	0.13	0.14	0.14
N	574	723	670

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

(b) Table IV: Robustness to Different Measures of Efficiency

Read: Table III is the core survival result. Net debt-to-capital has a negative and statistically significant effect on survival. In the baseline column, the marginal effect is approximately -0.180 . Return on sales and firm size raise survival, while labor cost lowers survival. Table IV replaces or supplements the efficiency controls with estimated inefficiency, cost per ton-mile, and 1980 return on sales.

Economic message: The negative leverage effect survives alternative measures of operating efficiency. This is central: the paper argues that debt is not merely standing in for inefficient operations. Financial structure has an independent predictive role for exit.

7.3 Table V: Effect of Leverage Controlling for the Ex Ante Risk of Default and Table VI: Summary Statistics Split According to the Industry Segment, Panels A and B

Table V
Effect of Leverage Controlling for the Ex Ante Risk of Default

The dependent variable is the probability that a general freight motor carrier with more than \$1 million in revenues in 1977 survives until 1985. The reported coefficients are probit estimates of the effect of a marginal change in the corresponding regressor on the probability of survival, computed at the average of the dependent variable. Net debt-to-capital is total debt minus cash reserves divided by total debt plus equity. Intangibles is the ratio of intangible assets to total assets. Labor cost is the ratio of the wages and benefits to operating costs. The first column reports the results of the basic regression of the probability of survival when the Altman Z-score is used as an additional explanatory variable. The Altman Z-score is an index, based on accounting information, of the risk of default. The higher the Z-score, the lower the probability of default is. The second regression reports results of the basic regression of the probability of survival conditioned on the fact the Z-score model predicted survival. The third regression reports the results of the basic regression of the probability of survival conditioned on the fact the Z-score model predicted default. All regressions contain nine regional dummies (not reported). Standard errors are reported in parentheses.

Independent Variables	Predicted Survivors	Predicted Defaults	
Net debt-to-capital	-0.150** (0.064)	-0.199** (0.078)	-0.183** (0.083)
Return on sales	2.344*** (0.397)	2.930*** (0.563)	2.138*** (0.526)
Log-revenues	0.039*** (0.015)	0.078*** (0.023)	0.017 (0.019)
Intangibles	-0.372 (0.262)	-0.639* (0.350)	0.107 (0.371)
Labor cost	-0.149 (0.150)	-0.331** (0.163)	-0.298 (0.320)
Z-score	0.012** (0.005)		
Pseudo- R^2	0.12	0.15	0.07
N	778	369	409

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

(a) Table V: Effect of Leverage Controlling for the Ex Ante Risk of Default

Table VI
Summary Statistics Split According to the Industry Segment

General freight carriers with more than \$1 million in operating revenues in 1977 are divided in three groups according to the fraction of their revenues coming from less-than-truckload (LTL) shipments as of 1977. Net debt-to-capital is total debt minus cash reserves divided by total debt plus equity. Intangibles is the ratio of intangible assets to total assets. Labor cost is the ratio of the wages and benefits to operating costs. Revenues per ton-mile is the ratio of operating revenues to ton-miles hauled. Average load is the ratio of ton-miles hauled to total miles traveled. Average haul is the ratio of ton-miles hauled to tons of revenues freight. Market share is the ratio between carriers' operating revenues and the sum of the operating revenues of all the carriers located in the same state. The Diff. column reports the difference in means between carriers that still existed in 1985 and carriers that did not exist at that date. The p -values refer to the null that the difference is equal to zero.

Variable	Mean	Median	Std. Dev.	Min.	Max.	Diff.	p -value	Obs
Panel A: Carriers with Less than 30 percent of revenues from LTL								
Revenues (\$ millions)	5.35	2.45	7.85	0.92	46.60	3.61	0.014	113
Assets (\$ millions)	2.06	0.90	3.32	0.16	23.70	1.86	0.003	113
Debt-to-capital	0.45	0.38	0.32	0.00	1.00	-0.03	0.575	113
Net debt-to-capital	0.27	0.25	0.43	-0.88	0.99	0.00	0.984	109
Coverage	22.27	7.52	30.10	0.00	100.00	3.52	0.538	112
Return on assets	0.19	0.19	0.13	-0.19	0.73	0.05	0.033	113
Return on sales	0.08	0.07	0.07	-0.03	0.42	0.02	0.059	113
Intangibles	0.07	0.03	0.08	0.00	0.44	-0.01	0.740	111
Labor cost	0.28	0.30	0.18	0.02	0.64	-0.04	0.198	112
Revenues per ton-mile	0.11	0.07	0.10	0.03	0.62	-0.01	0.625	90
Cost per ton-mile	0.19	0.07	0.41	0.03	3.68	-0.12	0.170	90
Average load	13.55	13.45	5.75	1.00	31.10	0.96	0.432	90
Average haul	252.51	217.64	196.32	6.93	1164.46	74.62	0.077	87
Market share	0.05	0.01	0.13	0.00	1.00	0.03	0.156	113
Number of employees	88.89	47.25	114.28	2.00	596.50	23.60	0.290	106
Panel B: Carriers with 30 to 70 percent of Revenues from LTL								
Revenues (\$ millions)	28.80	6.79	56.60	0.95	508.00	4.42	0.530	266
Assets (\$ millions)	13.80	2.48	32.00	0.05	309.00	5.06	0.203	266
Debt-to-capital	0.42	0.36	0.29	0.00	1.00	-0.13	0.000	265
Net debt-to-capital	0.25	0.26	0.36	-0.86	1.00	-0.20	0.000	257
Coverage	24.55	8.67	31.66	0.00	100.00	12.72	0.001	261
Return on assets	0.17	0.17	0.14	-0.30	0.88	0.08	0.000	264
Return on sales	0.07	0.07	0.07	-0.43	0.52	0.05	0.000	266
Intangibles	0.07	0.04	0.09	0.00	0.64	-0.01	0.293	265
Labor cost	0.52	0.55	0.13	0.01	0.76	-0.00	0.942	266
Revenues per ton-mile	0.24	0.18	0.21	0.06	1.86	0.03	0.230	259
Cost per ton-mile	0.26	0.18	0.32	0.06	2.46	0.04	0.286	259
Average load	10.90	11.45	4.21	0.80	29.54	0.35	0.504	259
Average haul	278.45	207.31	261.05	12.52	1713	-61.39	0.062	258
Market share	0.08	0.02	0.14	0.00	0.94	0.02	0.273	261
Number of employees	760.90	199.25	1421.08	2.00	13157	148.26	0.401	266

(b) Table VI: Summary Statistics Split According to the Industry Segment, Panels A and B

Read:Table V controls for Altman's Z-score and also splits firms by whether the Z-score predicts survival or default. Leverage remains negative and statistically significant. Table VI begins the TL/LTL split: Panel A covers carriers with less than 30% of revenue from LTL, and Panel B covers carriers with 30% to 70% of revenue from LTL.

Economic message: Table V weakens the interpretation that leverage only reflects ordinary bankruptcy risk. If leverage mattered only because highly indebted firms were already likely to default, controlling for Z-score should eliminate the result. It does not. Table VI sets up the key heterogeneity test: LTL carriers are larger, more network-intensive, and less purely competitive than TL carriers.

7.4 Table VI-Continued: Summary Statistics Split According to the Industry Segment, Panel C and Table VII: Differential Impact of the Initial Leverage on Survival in Different Segments of the Trucking Industry

Table VI—Continued

Variable	Mean	Median	Std. Dev.	Min.	Max.	Diff.	p-value	Obs
Panel C: Carriers with More than 70 percent of Revenues from LTL								
Revenues (\$ millions)	14.70	3.70	55.70	0.93	684.00	11.70	0.083	270
Assets (\$ millions)	6.73	1.45	28.50	0.23	355.00	6.83	0.048	270
Debt-to-capital	0.34	0.29	0.28	0.00	1.00	-0.15	0.000	270
Net debt-to-capital	0.14	0.09	0.37	-0.97	1.00	-0.20	0.000	261
Coverage	33.10	13.35	36.05	0.00	100.00	19.66	0.000	264
Return on assets	0.19	0.18	0.12	-0.33	0.56	0.06	0.000	270
Return on sales	0.08	0.07	0.05	-0.07	0.30	0.04	0.000	270
Intangibles	0.06	0.04	0.07	0.00	0.47	-0.02	0.007	268
Labor cost	0.56	0.59	0.11	0.02	0.78	0.00	0.923	269
Revenues per ton-mile	0.44	0.32	0.37	0.12	3.46	0.04	0.402	261
Cost per ton-mile	0.44	0.31	0.39	0.11	3.20	0.01	0.861	261
Average load	8.25	8.47	3.99	0.81	24.06	-0.35	0.477	261
Average haul	167.98	134.76	126.86	17.45	788.00	-11.60	0.461	261
Market share	0.05	0.01	0.12	0.00	0.90	0.02	0.204	270
Number of employees	426.64	114.25	1398.94	6.00	18015	310.07	0.072	264

(a) Table VI-Continued: Summary Statistics Split According to the Industry Segment, Panel C

Table VII

Differential Impact of the Initial Leverage on Survival in Different Segments of the Trucking Industry

The dependent variable is the probability that a general freight motor carrier with more than \$1 million in revenues in 1977 survives until 1985. The reported coefficients are probit estimates of the effect of a marginal change in the corresponding regressor on the probability of survival, computed at the average of the dependent variable. Net debt-to-capital is total debt minus cash reserves divided by total debt plus equity. Intangibles is the ratio of intangible assets to total assets. Labor cost is the ratio of the wages and benefits to operating costs. All the independent variables are measured as of 1977 (average of the 1976 and 1977 values). Among the independent variables there are also nine regional dummies (not reported). General freight carriers are divided into three groups according to the fraction of their revenues coming from less-than-truckload shipments as of 1977. The first column reports the estimates for motor carriers with less than 30 percent of revenues from less than truckload (LTL) shipments. Estimates in the second column are for motor carriers with more than 30 percent but less than 70 percent of their revenues in LTL shipments. Estimates in the third column are for motor carriers with more than 70 percent of their revenues in LTL shipments. Standard errors are reported in parentheses.

Independent Variables	LTL < 0.3	0.3 < LTL < 0.7	LTL > 0.7
Net debt-to-capital	0.021 (0.127)	-0.224** (0.102)	-0.277*** (0.101)
Return on sales	1.481* (0.829)	3.520*** (0.730)	3.247*** (0.821)
log(Revenues)	0.125* (0.064)	-0.010 (0.025)	0.063** (0.031)
Intangibles	0.151 (0.687)	0.089 (0.464)	-0.591 (0.512)
Labor cost	-0.417 (0.341)	0.086 (0.300)	-0.320 (0.341)
Pseudo- R^2	0.09	0.17	0.15
N	100	249	259

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

(b) Table VII: Differential Impact of the Initial Leverage on Survival in Different Segments of the Trucking Industry

Read:Table VI-continued reports the high-LTL group, carriers with more than 70% of revenues from LTL shipments. Table VII estimates the survival model separately by LTL share. Leverage has essentially no negative effect in the low-LTL group, but it is strongly negative in the middle- and high-LTL groups.

Economic message: The segment split is one of the paper's strongest pieces of evidence. The effect of debt appears where theory predicts it should matter most: in imperfectly competitive, network-intensive, service-sensitive LTL trucking. It is absent in the more competitive TL segment, where trucks are more collateralizable and service is more standardized.

7.5 Table VIII: Investments and Table IX: Firm-Level Prices and Leverage

Table VIII
Investments

This table reports the OLS estimates obtained by regressing capital investments of general freight motor carriers with more than \$1 million in revenues in 1977 on some predetermined explanatory variables. Investments are computed as differences in carriers' gross operating properties over the period divided by the 1977 level of net operating properties. Net debt-to-capital is total debt minus cash reserves divided by total debt plus equity. Intangibles is the ratio of intangible assets to total assets. Labor cost is the ratio of the wages and benefits to operating costs. All the independent variables are measured as of 1977 (average of the 1976 and 1977 values). The period 1977–1980 is also considered with the sample restricted to the motor carriers that survived until 1985 and with the sample restricted to motor carriers that exited between 1980 and 1985. All the regressions also contain nine regional dummies (not reported). Standard errors are reported in parentheses.

Independent Variables	Investment Period			
	1977–1985	1977–1980	1977–1980 of survivors	1977–1980 of nonsurvivors
Net debt-to-capital	-0.092 (0.203)	-0.224** (0.106)	-0.068 (0.131)	-0.379** (0.180)
Return on sales	2.249* (1.173)	2.208*** (0.693)	1.840*** (0.796)	1.619 (1.493)
log(Revenues)	-0.125** (0.052)	-0.063*** (0.028)	-0.057*** (0.033)	-0.034 (0.053)
Intangibles	-0.697 (1.137)	0.645 (0.518)	0.504 (0.733)	0.556 (0.761)
Labor cost	-1.081*** (0.393)	-0.087 (0.084)	-0.005 (0.082)	-1.438*** (0.400)
R^2	0.09	0.05	0.05	0.11
N	420	669	412	257

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

Table IX
Firm-Level Prices and Leverage

This table reports OLS coefficient estimates for a sample of general freight motor carriers with more than \$1 million in revenues in 1977 that disclose data on the volume of their shipments (Instruction 27 carriers). The dependent variable is the difference between the price per ton-mile charged by a carrier in the year of reference and the price charged in 1977. Net debt-to-capital in 1977 is debt minus cash reserves divided by debt plus equity. Labor cost in 1977 equals wages plus benefits over operating costs. A carrier market share in 1977 is the ratio between carriers' operating revenues and the sum of the operating revenues of all the carriers located in the same state. Medium less-than-truckload (LTL) carriers are carriers with more than 30 percent but less than 70 percent of their revenues in LTL shipments. High LTL carriers are carriers with more than 70 percent of their revenues in LTL shipments. All the regressions contain state fixed effects (not reported). Standard errors are reported in parentheses.

Independent Variables	1980	1981	1982	1983	1984	1985
Net debt-to-capital in 1977	0.046 (0.031)	-0.037 (0.040)	-0.138*** (0.051)	-0.126** (0.050)	-0.120** (0.058)	-0.103 (0.073)
Labor cost in 1977	0.092 (0.090)	0.088 (0.124)	0.087 (0.159)	0.249 (0.157)	0.328* (0.193)	0.201 (0.240)
Market share in 1977	0.063 (0.100)	0.121 (0.122)	0.058 (0.157)	0.138 (0.164)	0.101 (0.184)	0.102 (0.216)
Medium LTL revenues in 1977	0.054 (0.039)	0.089* (0.052)	0.063 (0.064)	-0.008 (0.065)	-0.015 (0.080)	-0.023 (0.101)
High LTL revenues in 1977	0.107** (0.042)	0.141** (0.055)	0.119* (0.068)	0.080 (0.069)	0.034 (0.082)	0.066 (0.106)
R^2	0.14	0.17	0.16	0.20	0.22	0.22
N	428	376	354	333	302	264

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

(a) Table VIII: Investments

(b) Table IX: Firm-Level Prices and Leverage

Read: Table VIII studies whether leverage affected real investment after deregulation. Initial leverage is especially negative for investment during 1977–1980 and for firms that eventually exit. Table IX studies price changes from 1980 to 1985 relative to 1977. Higher prederegulation leverage predicts lower prices in the recession and price-war years, especially 1982–1984.

Economic message: These tables provide mechanism evidence. Debt seems to limit investment when firms need to adapt to a new competitive environment. It also appears to push firms into lower pricing, possibly because they need cash or because customers require discounts to bear default risk.

7.6 Table X: Prices and Leverage in Different Segments of the Trucking Industry and Table A1: Production Function Estimate

Table X
Prices and Leverage in Different Segments of the Trucking Industry

This table reports OLS coefficient estimates for a sample of general freight motor carriers with more than \$1 million in revenues in 1977 that disclose data on the volume of their shipments (Instruction 27 carriers). The dependent variable is the vector of the differences between the price per ton-mile charged by a carrier in any year from 1980 to 1985 and the price charged in 1977. Net debt-to-capital in 1977 is debt minus cash reserves divided by debt plus equity. Labor cost in 1977 equals wages plus benefits over operating costs. A carrier market share in 1977 is the ratio between carriers' operating revenues and the sum of the operating revenues of all the carriers located in the same state. The first column reports the estimates for motor carriers with less than 30 percent of revenues from less than truckload (LTL) shipments. Estimates in the second column are for motor carriers with more than 30 percent but less than 70 percent of their revenues in LTL shipments. Estimates in the third column are for motor carriers with more than 70 percent of their revenues in LTL shipments. All the regressions contain state fixed effects and yearly dummies (not reported). Standard errors are reported in parentheses.

Independent Variables	LTL < 0.3	0.3 < LTL < 0.7	LTL > 0.7
Net debt-to-capital in 1977	0.087	0.027	0.017
*1980 dummy	(0.056)	(0.054)	(0.069)
Net debt-to-capital in 1977	0.108	-0.053	-0.080
*1981 dummy	(0.063)	(0.056)	(0.079)
Net debt-to-capital in 1977	0.080	-0.090	-0.239***
*1982 dummy	(0.062)	(0.058)	(0.079)
Net debt-to-capital in 1977	0.037	-0.064	-0.222***
*1983 dummy	(0.066)	(0.061)	(0.084)
Net debt-to-capital in 1977	0.002	-0.054	-0.212**
1984 dummy	(0.076)	(0.062)	(0.083)
Net debt-to-capital in 1977	0.074	-0.130**	-0.091
*1985 dummy	(0.079)	(0.066)	(0.091)
Labor cost	0.086	0.281***	0.185
	(0.083)	(0.083)	(0.144)
Market share	-0.078	0.189**	-0.112
	(0.514)	(0.076)	(0.134)
R^2	0.19	0.24	0.12
N	300	831	927

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level,

Table A1
Production Function Estimate

This table reports a fixed effect estimate of a translog production function where capital (K), labor (L), and intermediate goods (M) are the three factors. The sample is an unbalanced panel of firms present for at least two years in the period 1976 to 1978. The dependent variable is the logarithm of ton-miles hauled each year. Capital is the book value of net carrier operating properties. Labor is the number of workers employed. Intermediate goods are deflated expenses in fuel and supplies. As a factor controlling for possible heterogeneity of the principal input categories, the average haul and the average load plus two yearly dummies are used. The average load is the ratio of ton-miles hauled to total miles traveled. The average haul is the ratio of ton-miles hauled to tons of revenues freight. Standard errors are reported in parentheses.

$\log K$	-0.432***
	(0.078)
$\log L$	0.832***
	(0.112)
$\log M$	-0.334**
	(0.147)
$\frac{(\log K)^2}{2}$	0.040***
	(0.007)
$\frac{(\log L)^2}{2}$	0.110***
	(0.016)
$\frac{(\log M)^2}{2}$	0.052***
	(0.017)
$\log K \log L$	-0.043***
	(0.008)
$\log K \log M$	0.015**
	(0.007)
$\log L \log M$	-0.045***
	(0.012)
Average load	0.049***
	(0.002)
Average haul	0.002***
	(0.000)
1977 dummy	0.014**
	(0.006)
1978 dummy	0.000
	(0.007)
R^2	0.60
N	1683

***, **, * Significantly different from zero at the 1 percent level or less, at the 5 percent level, or at the 10 percent level, respectively.

(a) Table X: Prices and Leverage in Different Segments of the Trucking Industry

(b) Table A1: Production Function Estimate

Read: Table X repeats the pricing analysis by trucking segment. The strongest negative pricing effects of leverage appear in the high-LTL segment, especially in 1982–1984. Table A1 reports the translog production-function estimates used to construct the technical inefficiency measure in robustness checks.

Economic message: Table X lines up with the survival heterogeneity in Table VII. Debt matters most for pricing where service, reputation, and imperfect competition are most important. Table A1 supports the robustness exercise showing that the leverage result survives when efficiency is measured using a production-frontier approach rather than accounting profitability alone.

8 Main findings by theme

8.1 Finding 1: efficient firms are more likely to survive

- Return on sales and return on assets predict survival positively.
- Larger firms are more likely to survive.
- Higher labor cost shares, which partly proxy for union exposure and less use of purchased transportation, reduce survival.

- These patterns are consistent with the standard “survival of the fittest” view.

8.2 Finding 2: leverage reduces survival, beyond efficiency

- Net debt-to-capital has a robust negative effect on the probability of survival.
- The result remains after using alternative efficiency measures and after controlling for Altman’s Z-score.
- Economically, the paper reports that a one-standard-deviation increase in leverage reduces the probability of survival by about 8 percentage points.
- A carrier would need meaningfully higher profitability to offset the survival penalty associated with a higher leverage ratio.

8.3 Finding 3: debt matters most in LTL trucking

- The low-LTL group behaves like a competitive, asset-collateralizable TL segment: leverage has no significant negative effect on survival.
- In medium- and high-LTL groups, leverage sharply reduces survival probability.
- This pattern is consistent with debt overhang, predation, reputation, and customer-confidence channels.
- It is less consistent with a story in which leverage simply captures unobserved poor quality common across all segments.

8.4 Finding 4: leverage constrains investment

- Leverage is negatively related to investment during 1977–1980.
- The effect is especially strong for carriers that later exit.
- This suggests that high debt may prevent struggling carriers from making the investments needed to adapt to deregulation.

8.5 Finding 5: leverage affects prices

- Highly leveraged carriers charged lower prices in the most intense postderegulation price-war years.
- The price effect is strongest in the LTL segment, where quality and continuity of service matter more.
- The paper interprets this as evidence that leverage affects competitive behavior, not only bankruptcy probabilities.

9 Identification and interpretation

9.1 Why deregulation helps identify financing effects

- Capital structure is normally endogenous to strategy, risk, and product-market conditions.

- The deregulation shock was largely unexpected and changed the industry environment after leverage had already been chosen.
- Deregulation also reduced the value of operating certificates, mechanically increasing effective leverage.
- The shock therefore pushes observed leverage away from the leverage that firms would have chosen for the deregulated environment.

9.2 Remaining identification concerns

- **Unobserved quality.** Highly levered carriers may have had worse unobserved prospects even before deregulation.
- **Measurement of efficiency.** Accounting profitability may reflect monopoly rents as well as operating efficiency.
- **Exit measurement.** Some firms disappear because they are acquired and merged rather than liquidated.
- **Private-firm data limitations.** Most firms are privately held, which limits market-value measures and makes Tobin's q unavailable.

9.3 How the paper responds

- It controls for multiple measures of efficiency: ROS, ROA, production-frontier inefficiency, cost per ton-mile, and 1980 ROS.
- It controls for Altman's Z-score to separate leverage from ex ante default probability.
- It studies mechanisms, showing leverage predicts investment and pricing behavior.
- It uses TL/LTL heterogeneity as a cross-market check. The leverage effect appears where theory predicts product-market finance channels should be strongest.

10 Economic intuition

10.1 Why did debt hurt carriers?

Debt is costly in the paper because the deregulation shock increased the need for flexibility at exactly the time when highly levered firms had weak balance sheets. A firm with little debt could absorb losses, invest in adaptation, and price strategically without immediately risking insolvency. A highly levered firm faced tighter financing constraints and greater pressure to generate cash.

10.2 Why did debt matter more in LTL?

LTL service relies on networks and reliability. Customers may care about the possibility that the carrier fails mid-shipment. Competitors may be able to exploit weak rivals. Investment needs are less purely collateralizable than trucks alone. These features make financial weakness strategically important. In contrast, TL is closer to a commodity-like service with lower entry barriers and more tangible assets.

10.3 Survival of the fittest versus fattest

The paper does not reject the idea that efficient firms survive. Rather, it shows that efficiency is not the whole story. In a world with capital-market frictions, survival depends on both operating efficiency and financial capacity. Deregulation selected firms by both productivity and balance-sheet strength.

11 Relation to broader literature

- **Debt overhang.** The evidence is consistent with Myers-style underinvestment: debt discourages investment when the gains would accrue partly to creditors.
- **Product-market finance.** The results complement theories in which financing affects competitive behavior, such as predation and strategic interaction.
- **LBO evidence.** The paper complements studies of leveraged buyouts by using an industry deregulation shock rather than leverage transactions chosen by firms.
- **Financial constraints.** The paper connects firm-level finance to industry dynamics by showing that external finance frictions can shape the composition of survivors.

12 Short summary

- The paper studies U.S. trucking deregulation as a natural experiment to test whether financing affects survival.
- The main sample is general freight carriers with more than \$1 million in 1977 revenues, followed from 1976 to 1985.
- Highly levered firms are less likely to survive, even conditional on profitability, size, labor costs, intangibles, region, production-frontier inefficiency, cost per ton-mile, 1980 profitability, and Altman's Z-score.
- The effect of leverage is concentrated in LTL trucking, the less competitive and more network-intensive segment.
- High leverage is associated with lower investment early in the deregulation period, especially among eventual exiters.
- High leverage is also associated with lower prices during the price war, especially in high-LTL carriers.
- The core message is that industry shakeouts do not necessarily select only the most efficient firms; they can also select firms with stronger financial capacity.

13 Conclusion

13.1 Core conclusion

Zingales (1998) shows that capital structure can affect real outcomes in product markets. In the trucking industry, financially weak carriers were less likely to survive deregulation, even when their

observed operating characteristics were similar to those of less levered carriers.

13.2 Incremental contribution revisited

The paper’s key contribution is to use an unexpected deregulation shock to study a corporate-finance question that is otherwise difficult to identify. Because firms chose leverage under the regulated regime, prederegulation leverage is less likely to be an optimal response to the deregulated competitive environment. This gives the survival results a causal interpretation that would be hard to obtain in a normal cross-section.

13.3 Economic implications

- **For industrial organization.** Exit after deregulation reflects financial strength as well as productivity.
- **For corporate finance.** Debt can shape investment, pricing, and survival, especially in imperfectly competitive sectors.
- **For policy.** Deregulation and other competitive shocks may have distributional effects across firms depending on balance sheets, not just efficiency.
- **For empirical work.** Natural experiments that shock product markets can help identify the real effects of financing choices.

13.4 Limitations

- The data are based on regulatory filings and are limited for smaller and privately held firms.
- Market-value measures of assets, investment opportunities, and Tobin’s q are mostly unavailable.
- The exit measure combines liquidation, bankruptcy, and merger into another carrier.
- The natural experiment is powerful, but leverage may still be correlated with unobserved firm quality.
- The setting is a specific regulated industry experiencing a large shock, so external validity should be judged carefully.

13.5 Takeaway

The clean takeaway is the title’s contrast: after deregulation, trucking survival was not simply “survival of the fittest.” It was also partly “survival of the fattest.” Efficient carriers were more likely to survive, but firms with stronger balance sheets were also more likely to withstand the competitive shock.